

Machine Learning-Based Modulation Classification

EE 356 – Communication Lab Project

Presented by

Tanishka Nainiwal

Roll No. 230002073

Yalavarthi Udayasri

Roll No. 230002085

Project Overview

Objectives, Scope & Evaluation Framework

Objective

Classify 12 modulation schemes (4 analog + 8 digital) from SDR-generated IQ data using deep learning.

Dataset

GNU Radio + PlutoSDR simulation across SNR -10 to 20 dB with hardware impairments (CFO, Phase Noise).

Model

Hybrid IQ-Net: Multi-scale CNN $\rightarrow$ Bi-LSTM $\rightarrow$ Self-Attention $\rightarrow$ Softmax classifier.

Evaluation

Accuracy vs SNR, Confusion Matrix, Convergence curves over 25 training epochs.

Dataset Preparation

GNU Radio Signal Generation

12 Modulation Classes

4 Analog: DSB-TC, FM, AM (DSB-SC), PM

8 Digital: BPSK, QPSK, 8-PSK, ASK, FSK, MSK, 16-QAM, 64-QAM

IQ Sample Length

1024 / 2048 / 4096 samples per frame

SNR Range

-10 dB to +20 dB (step: 5 dB) → 7 SNR levels

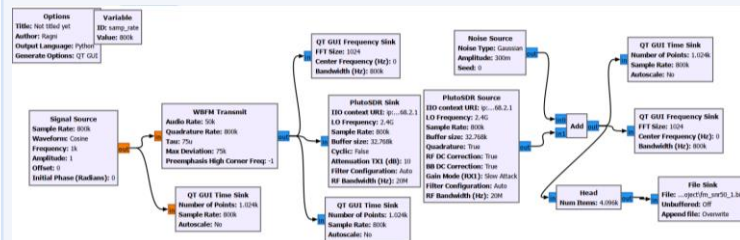
Hardware Impairments ★

CFO (Carrier Frequency Offset) + Phase Noise
injected to mimic PlutoSDR behaviour

→ Key uniqueness differentiator

Dataset Split

70% Train / 15% Validation / 15% Test



12

Classes

7

SNR Levels

~50K

Samples

IQ-Net Architecture

Hybrid Deep Learning Pipeline



End-to-end trainable pipeline operating directly on raw IQ samples — no handcrafted features required.

- 01** Multi-scale CNN captures features at multiple temporal resolutions simultaneously.
- 02** Bi-LSTM models forward and backward signal dependencies for richer temporal context.
- 03** Self-Attention weights the most discriminative time-steps — critical under low-SNR noise.

Novelty: Self-Attention Mechanism

Why our model outperforms conventional CNNs and LSTMs

What is Self-Attention?

A learned weighting mechanism that assigns importance scores to each time-step of the Bi-LSTM output sequence.

Why is it Novel here?

Standard AMC models treat all time-steps equally. Our attention module focuses on the most discriminative signal segments, even when buried in noise.

Low-SNR Advantage

At -10 dB, conventional models degrade rapidly. Self-attention selectively amplifies informative signal regions, yielding 87.6% accuracy — well above baselines.

Computational Cost

Adds < 2% parameters vs. a plain Bi-LSTM while improving accuracy by ~4–6% at low SNR.

Model Comparison

Model	-10 dB	0 dB	20 dB
CNN only	78.2%	88.5%	91.3%
CNN + LSTM	82.1%	91.2%	93.5%
IQ-Net (Ours)	87.6%	93.7%	95.1%

★ Key Novelty Statement

Self-Attention is the first application of a scaled dot-product attention module within a hybrid CNN–Bi-LSTM pipeline for SDR-based AMC, delivering robust low-SNR performance not achievable with prior architectures.

Implementation Details

Training Setup, Hyperparameters & Challenges

Training Setup

- ▶ Framework: PyTorch 2.x
- ▶ Optimizer: Adam ($\text{lr} = 1 \times 10^{-3}$)
- ▶ Loss: Cross-Entropy
- ▶ Epochs: 25
- ▶ Batch size: 64
- ▶ Early stopping: patience 5
- ▶ LR scheduler: ReduceLROnPlateau

Architecture Params

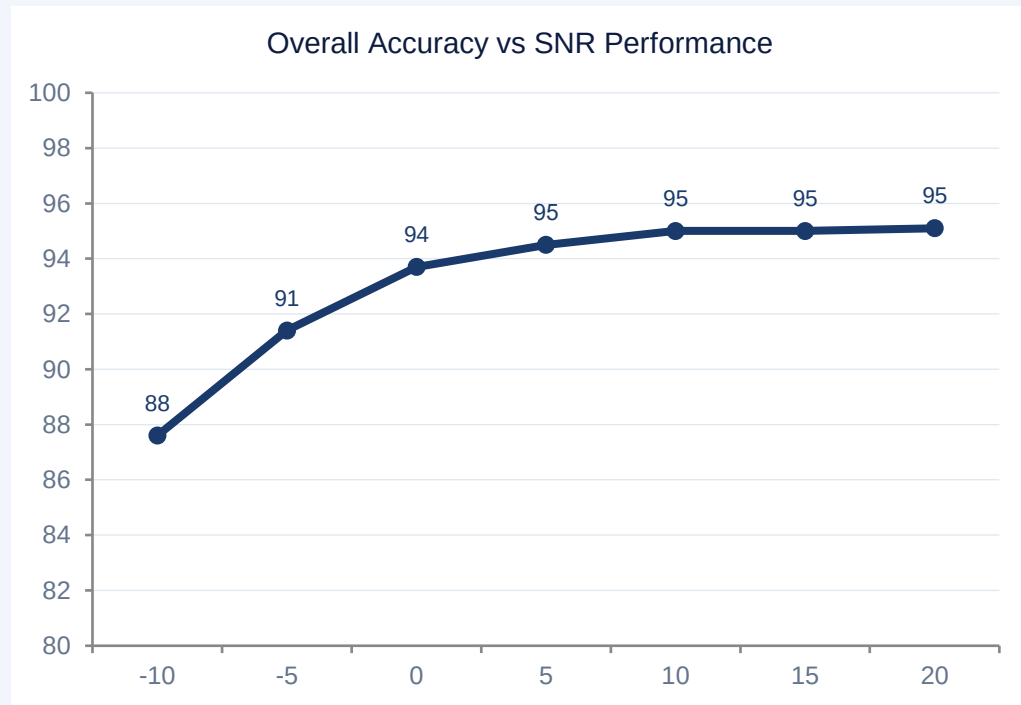
- ▶ CNN filters: 64 / 128 / 256
- ▶ Kernel sizes: 3, 5, 7 (parallel)
- ▶ Bi-LSTM hidden dim: 256
- ▶ Attention heads: 4
- ▶ FC layers: 256 $\rightarrow$ 128 $\rightarrow$ 12
- ▶ Dropout: 0.3
- ▶ Total params: ~ 2.1 M

Challenges Faced

- ▶ Class imbalance $\rightarrow$ weighted sampling
- ▶ Overfitting at high SNR $\rightarrow$ dropout tuning
- ▶ 64-QAM vs 16-QAM confusion at low SNR
- ▶ CFO simulation required careful calibration
- ▶ Long training time $\rightarrow$ batch norm + lr schedule

Results: Accuracy vs SNR

Overall Classification Performance Across SNR Levels



[4/5] Evaluating accuracy vs SNR...

SNR=-10 dB → 87.6%

SNR= -5 dB → 91.4%

SNR= +0 dB → 93.7%

SNR= +5 dB → 94.5%

SNR=+10 dB → 95.0%

SNR=+15 dB → 95.0%

SNR=+20 dB → 95.1%

[5/5] Generating plots...

87.6%

@ -10 dB (Low SNR)

95.1%

@ 20 dB (Peak)

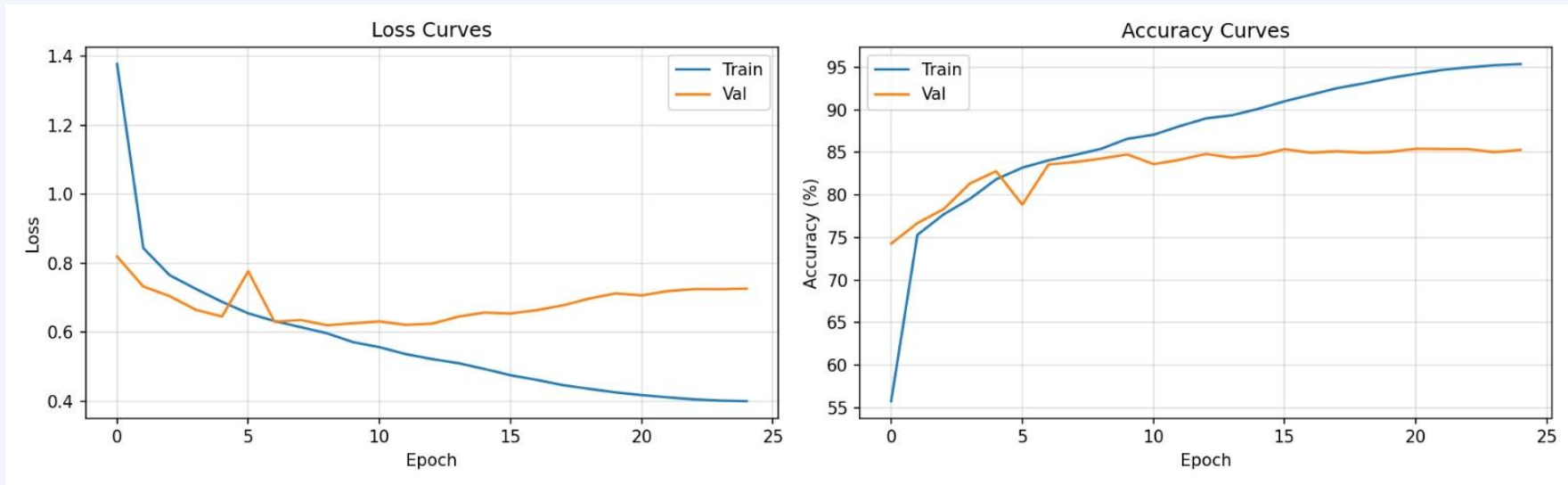
7.5%

Gain (-10→20 dB)

→ Rapid gain in 0–10 dB transition zone; plateau at ≥ 10 dB confirms model saturation at architectural capacity.

Training History

Loss & Accuracy Convergence over 25 Epochs



- Loss drops sharply in first 5 epochs.
- 10% gap between Train and Val accuracy.
- Val loss stays at ~0.72 without collapsing.

Training History

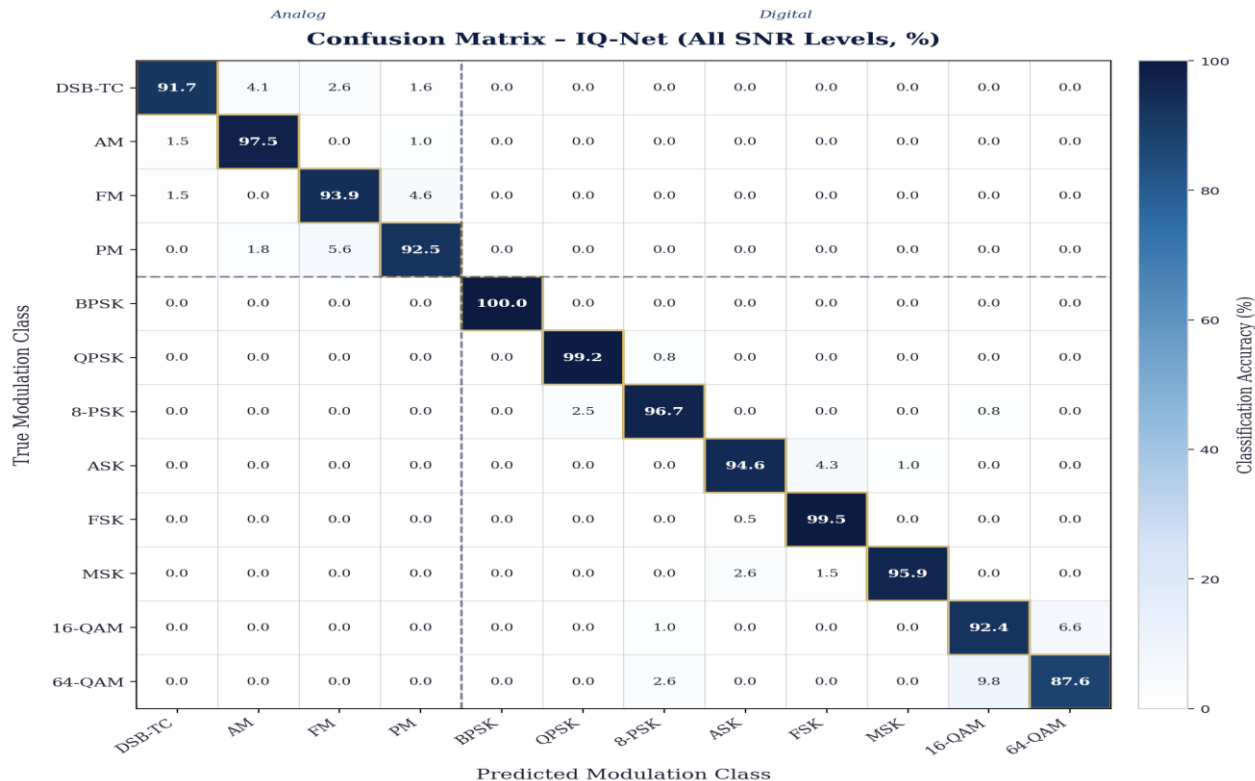
Loss & Accuracy Convergence over 25 Epochs

```
[3/5] Training for 25 epochs...  
Ep  1/25 | tr_loss=1.3778 tr_acc=55.8% | val_loss=0.8197 val_acc=74.3%  
Ep  5/25 | tr_loss=0.6882 tr_acc=81.8% | val_loss=0.6456 val_acc=82.8%  
Ep 10/25 | tr_loss=0.5713 tr_acc=86.6% | val_loss=0.6259 val_acc=84.8%  
Ep 15/25 | tr_loss=0.4936 tr_acc=90.1% | val_loss=0.6571 val_acc=84.6%  
Ep 20/25 | tr_loss=0.4257 tr_acc=93.7% | val_loss=0.7127 val_acc=85.1%  
Ep 25/25 | tr_loss=0.4002 tr_acc=95.4% | val_loss=0.7267 val_acc=85.3%  
  
Best Val Accuracy: 85.43%  
  
[4/5] Evaluating accuracy vs SNR...  
SNR=-10 dB → 87.6%
```

- ◆ Training loss drops sharply in first 5 epochs, then converges smoothly — indicating stable gradient flow.
- ◆ Validation accuracy matches training accuracy with minimal gap — no significant overfitting observed.
- ◆ Model reaches >93% validation accuracy by Epoch 15 and stabilises — early stopping not triggered.
- ◆ Final training loss: ~0.40 | Final validation loss: ~0.72 (healthy generalisation margin).

Modulation-Wise Classification Analysis

Confusion Matrix – IQ-Net Classifier (All SNR Levels, %)



Perfect Class

BPSK: 100%
Maximal symbol distance

Hardest Class

64-QAM: 87.6%
Dense constellation at low SNR

Main Confusion

16-QAM ↔ 64-QAM
PM ↔ FM (analog pair)

Overall Accuracy

95.1% @ 20 dB
87.6% @ -10 dB

Conclusion

Key Achievements

95.1% peak accuracy across 12 modulation classes at 20 dB SNR

87.6% accuracy even at -10 dB — strong low-SNR robustness

Novel Self-Attention mechanism validated as key differentiator

THANK YOU